Smart Assistant for Senior Citizens

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A PROJECT SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR OF SCIENCE (COMPUTER SCIENCE)

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DECLARATION

We, the candidates of Bachelor of Science (Computer Science) at Shaheed Zulfikar Ali Bhutto Institute of Science and Technology, Islamabad do hereby certify that this report titled **Text to BPN Process and Decision Engine**, submitted as partial fulfillment of Bachelor of Science (Computer Science) degree requirements, is our original work and we are its sole author. All the employed materials, references to the literature and the work of others have been referred to and duly cited. This report has not been presented for examination anywhere else.

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Project Overview

The global aging population presents a significant challenge to modern healthcare systems, necessitating innovative technologies that promote "aging in place". Currently, senior citizens face profound challenges in maintaining independence due to physical mobility issues and the inherent complexity of managing modern healthcare routines. There is a critical lack of integrated systems that combine continuous health monitoring with active social support, often leading to undetected domestic accidents and life-threatening medication non-compliance. Social isolation further exacerbates these issues, as many elderly individuals live alone without professional care or a reliable mechanism to seek help for simple daily tasks. This disconnection creates a high-risk environment characterized by higher rates of preventable hospitalizations and a significant decline in mental health.

The Proposed Solution is a comprehensive digital ecosystem designed to facilitate the independence of the elderly through the integration of modern assistive technologies. This web-based system bridges the critical gap between seniors, their families, and a verified network of community volunteers. By combining artificial intelligence, IoT-based health tracking, and real-time communication protocols, the project establishes a secure and proactive environment for aging populations. The system directly addresses the problem of undetected accidents by implementing AI-driven fall detection and automated SOS broadcasting to a prioritized contact hierarchy. Ultimately, the platform transforms traditional caregiving into a data-driven, community-supported model that ensures safety without compromising user autonomy.

The system provides a unified interface for health oversight, emergency coordination, and proactive wellness management. Its core functionality is divided into specialized modules that handle specific aspects of care: Emergency and Safety: Features AI fall detection using LSTM networks and a one-tap SOS button. **Health and Medication:** Provides real time vital sign tracking via IoT sensors and automated medication reminders. **Task Request and Volunteer Matching:** Utilizes AI to match elderly users with nearby volunteers for physical tasks based on location and skill. **Adaptive Assistance:** Prioritizes accessibility through voice command navigation, gesture control, and a computer vision object identifier. **Smart Travel Escort:** Implements geofencing technology to monitor route deviations and provide navigation guidance.

The implementation of this system significantly improves the sense of safety and independence among elderly users, allowing them to manage daily routines with greater confidence. Caregivers and family members benefit from reduced psychological stress and burnout, as the platform provides real-time data and reliable community assistance. Furthermore, the project benefits the broader community by creating a scalable framework that optimizes how volunteers interact with the vulnerable elderly population. By shortening emergency response times and minimizing health-related risks through consistent monitoring, the platform builds a safer, more connected, and compassionate ecosystem for elderly support.

Dedication

I dedicated this work to my beloved parents and family, who have been my greatest source of strength and inspiration throughout this academic journey. Their unwavering belief in my potential and the countless sacrifices they made to provide me with a quality education at SZABIST University have made this achievement possible. This project is a tribute to their love and the values of compassion and service they have instilled in me.

I also wish to dedicate this project to my supervisor, Mr. Ghaffar Ahmed, whose professional guidance and mentorship were instrumental in shaping this smart assistant. His constant encouragement and technical insights provided the necessary foundation to transform this vision into a reality for the benefit of senior citizens.

Acknowledgment

I wish to express my deepest gratitude to my supervisor, Mr. Ghaffar Ahmed, for his invaluable guidance and constant support throughout the development of this project. He provided critical feedback during the compilation of all twelve sections of this proposal, ensuring the technical feasibility of the AI and IoT integrations. I am truly grateful for his patience and the expertise he shared, which helped refine the core modules of the Smart Assistant. His mentorship has been instrumental in aligning this project with modern computing standards.

I also extend my thanks to the faculty members of the Department of Computer Science at SZABIST University for their academic contributions. The technical foundations provided by our instructors allowed us to implement complex features such as LSTM networks for fall detection and MongoDB for scalable data management. I am also thankful to the peer reviewers who examined the application comparisons, helping us identify the operational gaps in existing platforms. Their insights were vital in defining the unique capabilities of our proposed ecosystem.

I would like to recognize the institutional support provided by SZABIST University, Islamabad Campus. The Faculty of Computing and Engineering Sciences provided the necessary environment and resources to conduct this research during the Spring 2026 semester. I am particularly grateful for the access to the digital repositories and laboratory facilities that enabled the exploration of IoT health telemetry and geospatial indexing techniques. The university's commitment to innovation in digital health has been a primary driver for this work.

Finally, I want to thank my teammates for their dedication and collaborative spirit throughout the project's revision history. I am also deeply appreciative of my family and friends, whose encouragement provided the emotional resilience needed to complete this Final Year Project. Their belief in the significance of this work, which aims to improve the quality of life for the elderly, motivated us to strive for excellence at every stage.

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Chapter 1

Introduction

The modern healthcare geography presently faces a massive shift as the growing population grows at an unknown rate, putting heavy pressure on traditional care systems and family support structures. To address these critical scarcities, this exploration proposes the development of the project, a comprehensive technological ecosystem combining a point-rich web platform with an integrated IoT tackle. By polarizing functions similar to real-time physiological monitoring, automated fall discovery, and AI-driven activity tracking, the operation provides a constant safety net for the senior. Beyond clinical safety, the design introduces a unique community-driven approach that facilitates social engagement through a localized levy network and a devoted price frame. The perpetuation of this system carries profound significance by offering a scalable result that enhances individual autonomy through voice- enabled navigation and accessible design. Families profit from a reduced cerebral burden, while the integration of constant heart rate telemetry and geofencing ensures that exigency response times are minimized. Operationally, the system functions as an intelligent conciliator, exercising a suite of connected modules to automate health compliance and social collaboration. Eventually, the System effectively closes the gap between fractured shadowing tools and the complex everyday requirements of aging. By streamlining communication between seniors, caregivers, and levies, the proposed design creates an inclusive, data- driven support network that elevates the quality of life for a vulnerable demographic.

1.1 Product Purpose

The proposed system is a website that keeps users safe while assisting them with daily tasks. Elderly people can ask for help and speak with those who care for them here. It includes items that monitor people's health. Using it enables people to communicate with each other as necessary. It attempts to address issues such as when people are alone or do not receive assistance when they require it. contains features like medication reminders and buttons that users can push in the event of a fall. It includes a mechanism to locate volunteers who are close to the individuals and can assist them. It also sends messages to those who look after the people to let them know what is happening. In this manner, users are still able to perform tasks independently. Be secure and help elderly people find helpers. People's lives are improved by the system because it provides a means of safety and dependable elderly people.

1.2 Product Scope

In software engineering, the project scope outlines the precise features and workload needed to complete a development project, defining its boundaries. It fosters a common understanding among all stakeholders by serving as a comprehensive guide that makes a distinction between duties that are included and those that are not. The team controls expectations and avoids scope creep concerns by setting these boundaries. This clarity guarantees the effective use of technical and human resources throughout the lifecycle.

In the end, a clearly defined scope offers the crucial starting point required to confirm that the finished product complies with its initial requirements[1].

The following sections describe the particular modules that make up the project along with the main features linked to each module. This comprehensive analysis outlines the specific features the platform will provide to enhance elderly autonomy

User Authentication and Login

- Secure Login or signup
- One Time Password email or Password
- Role selection for elder, volunteer, and caregiver
- Account recovery and verification
- Account deletion

User Profile Management

- Granular Privacy Controls
- Emergency Contact Hierarchy
- Activity and Interaction Logs
- Digital Document Vault
- User Profile Customization

Task Request Module

- AI Task Prioritization
- Smart Volunteer Matching
- Voice-to-text request
- Live Task Tracking
- Proof of Completion

Communication

- Multi-Channel Calling
- Integrated Messaging
- Schedule call reminders
- Group and Family Chats
- Real-time SOS broadcasting

Health and Medication

- Medication Reminders
- Digital Health Repository
- Appointment Management
- IoT Health Telemetry

- Drug Interaction Engine

Wellness and Physical Rehab

- AI-Guided Physiotherapy
- Meal and Nutrition Tracking
- Interactive Audio Coaching
- Real-Time Pose Correction
- Fitness Goal tracking and Sleep timer

Emergency and Safety

- AI fall detection
- Multi-Modal Distress Analytics
- Biometric Stress Trigger
- Temporal Inactivity Gate
- One-tap SOS button

Social Circle

- Generate Auto-replies
- Event discovery
- Hobby groups and social forums
- Community announcements
- Virtual group activities

Volunteer Rewards

- Points and badges
- Competitive Leaderboards
- Credentialing System
- Performance Analytics
- Professional Skill Validation

Smart Travel Escort

- Context-Aware Route Guidance
- Companion Booking System
- Geo-Fencing and Deviation Alerts
- Mobility-Matched Transport
- Ride-Hailing Integration

Adaptive Assistance

- Voice commands navigation
- Adaptive Accessibility UI
- Computer Vision Object Identifier
- IMU-Based Gesture Control
- Multi-Modal Screen Reading

Emergency Guide

- Tactical Emergency Checklist
- Post incident reporting and feedback
- Context-Aware Incident Handling
- Proximity-Gated Responder Alerts
- Incident-Specific Audio Guidance

1.3 Objectives

In software engineering, objectives serve as measurable standards that guide the course of a project's development. These goals convert vague ideas into precise, high-fidelity technical outcomes that are essential for deployment success. An engineering project creates a clear plan that enables the meticulous tracking of milestones by defining these metrics. Additionally, well-defined parameters improve transparency among all project participants and maximize resource distribution. In the end, this methodical approach guarantees that the finished system satisfies its functional criteria and accomplishes overall mission success[2].

The proposed project functions as a holistic platform with integrated software and hardware solutions to improve elderly citizens' safety, health, and social connectivity. Fundamentally, the architecture ensures that specialized roles for elders, volunteers, and caregivers operate under stringent privacy constraints by using an OTP-verified registration process and a digital document vault. A key element of this project is a high-performance wearable that uses the MAX32664 sensor to continuously track heart rate and oxygen levels biometrically. It is driven by the SparkFun ESP32. The system uses sophisticated TDK InvenSense motion sensors to allow automatic fall detection and distress analysis to optimize physical safety. The usefulness is further expanded by a wellness package that includes interactive nutrition tracking, a customized engine to prevent medication errors, and Mediapipe for skeletal tracking during their physical therapy. Dedicated modules that support family group conversations, hobby groups, and community event integration are used to treat social isolation. A task matching engine, which uses a scoreboard and reward system to promote community involvement, links elders with certified volunteers for operational support. A travel escort function that integrates ride-sharing, geofencing, and computer-vision capabilities to help individuals with limited vision recognize home objects improves mobility. With voice-command navigation and flexible screen layouts, accessibility is still a top emphasis. Lastly, the system offers a centralized dashboard for thorough long term health and activity monitoring and guarantees prompt crisis assistance with a tactical emergency guide and a one tap SOS button.

Achieving these specific goals will ensure that the final product aligns with its purpose of providing a dependable safety net for the elderly. By creating an effective data pipeline

that connects hardware to the cloud, the project will provide a health monitoring solution that adheres to the high safety and accuracy standards essential in the senior care sector. Accomplishing the objectives for multi-modal detection and interaction will greatly enhance the speed of emergency response, contributing to the overarching vision of preserving independence and dignity for senior citizens. Additionally, incorporating features for physical therapy, nutrition, and social interaction will transform the product from a basic alert system into a comprehensive wellness platform. In the end, these technical achievements will meet the requirements of both the primary users and their caregivers.by creating a smart, data driven support network. These results will ensure long term peace of mind through constant technological innovation and user focused design.

1.4 Intended Market of Product

The proposeed system is intended to help a number of different markets that work with and care for older people. These are healthcare providers, families and caregivers, and groups of volunteers in the community. Each of these markets has its own problems and needs when it comes to keeping seniors healthy. These sections outline each target market and explain how the system meets their needs and gives them valuable benefits.

Healthcare professionals: They providers dedicated to geriatric care, represent a critical market segment focused on continuous monitoring and emergency intervention. Currently, these providers face significant challenges, such as the inability to track patients in real time and delays in receiving critical emergency alerts outside of clinical settings. The Smart Assistant for Senior Citizens directly addresses these gaps by integrating IoT-based health monitoring with AI-driven safety features. The system continuously tracks vital signs, manages medication adherence, and utilizes automated fall detection to ensure healthcare workers can monitor patients remotely and intervene immediately during medical crises. By maintaining comprehensive digital health records, the platform enables data-driven decision-making for more accurate diagnoses. In the future, this technology will empower providers to manage patient populations more effectively while reducing the overall clinical workload for hospitals. Early detection of health deviations will lead to superior treatment outcomes and improved patient longevity. Ultimately, the system will enhance the quality of care, lower operational costs, and significantly accelerate emergency response times across the healthcare landscape.

Families and Caregivers Families and caregivers are the people who take care of and watch over older people every day. This market is made for caregivers who often have a hard time keeping an eye on older people all the time. Right now, they have problems like worrying about safety, not getting real-time updates, and having trouble handling emergencies from far away. ALll of these problems are solved by sending real-time alerts, tracking health, and sending emergency alerts in this platform. There are a few features that keep seniors safe when they are alone like AI-based fall detection, SOS alerts, and medication reminders. Caregivers will also stay updated and connected using the built in tools. Caregivers will have less stress and more confidence in the future when they have to take care of elderly people from a distance. It will help seniors and their families talk to each other more. The platform will also save time and effort by taking care of routine monitoring tasks on its own. This market will benefit from better care quality, more peace of mind, and better coordination.

Community Services and Volunteer Community service groups and volunteer organisations work to help older people with their daily lives. This market has non-profits and local volunteers who want to help older people feel better. Right now, these groups have problems like poor coordination, no structured systems, and trouble finding seniors who need help. The Smart Assistant for Senior Citizens is useful for this market because it adds a task request module that smartly matches volunteers based on where they are and when they are available. It makes it easy for seniors to ask for help and puts them in touch with verified volunteers. The platform also has features for tracking and communication that will make sure that all the tasks are finished. In the future volunteer organizations will be able to serve more seniors in less time and manage tasks more effectively. It will decrease assistance delays and enhance coordination. Additionally, the system's monitoring and verification features will foster trust. In general, this market will gain from enhanced service delivery, increased community involvement, and better resource utilization.

1.5 Intended Users of the Product

The proposed digital ecosystem is intended for various types of users who will engage with the system to establish a coordinated care setting. The descriptions detail each user type, including an explanation of their roles and the advantages they will gain from utilizing the system. This acts as a guide for comprehending how various stakeholders work together on the platform to guarantee the safety and independence of seniors.

Senior Citizens: These individuals are the main users of the system, frequently encountering difficulties like physical mobility problems, social isolation, and the complications of handling chronic health regimens. At present, numerous individuals live independently and face challenges in securing dependable help for everyday activities such as interpreting prescriptions or traveling unfamiliar paths. As they start utilizing the product, they will engage with the Adaptive Assistance module via voice commands and gesture-controlled interfaces. They will use the Task Request Module to ask for assistance from volunteers and depend on the Emergency and Safety module for automatic fall recognition. This platform will offer them a substantial reclamation of autonomy and an increased feeling of security. They will gain from quicker emergency response times in medical emergencies and a decrease in the cognitive deterioration linked to social isolation

Caregivers and Family Members: These individuals are accountable for the distant supervision of their aging family members, frequently facing significant mental strain and exhaustion because of insufficient real-time information. In today's environment, they frequently depend on disjointed communication and manual updates that do not offer a cohesive safety net. By adopting this system, they will employ the Caregiver Dashboard to oversee real-time health data and geofenced route deviations. They will engage with the Health and Medication module to monitor medication adherence and get instant SOS alerts in case of an accident. The product will greatly alleviate their emotional and financial strain by offering a proactive, data-informed monitoring solution. They will feel reassured knowing that assistance is sent out through a prioritized system, even in their absence

Community Volunteers: These users are community members eager to help seniors with physical activities, but they presently do not have an organized digital platform

to recognize and address needs effectively. Currently, their input is frequently informal and overlooked, hindering the ability to maintain a dependable support system. With the suggested product, volunteers will utilize the Task Request Module to get alerts based on proximity for assistance requests. They will engage with the Volunteer Rewards module, allowing them to monitor their performance analytics and obtain validation for their professional skills. The platform will offer them a structured system that acknowledges and compensates their efforts with points, badges, and competitive rankings. This will lead to a more involved and caring community environment that enhances the delivery of localized support to the at-risk elderly demographic

Chapter 2

Background and Literature Review

The primary purpose of a literature review aims to provide a solid foundation for the proposed system. It assesses current technologies by outlining their pros and cons within geriatric digital health. The review identifies major knowledge gaps, such as the absence of proactive emergency detection and community support. It addresses inefficiencies in fragmented care models, ensuring that development is based on proven methods. This chapter analyzes relevant literature and platforms to confirm the project's benefits for senior independence and supports a transition to a validated solution [3].

This chapter serves as a guide for the sections that follow. It begins with Section 2.1, which looks at existing systems to see what they can do and where they fail. Section 2.2 then describes the potential usage of a new system to show how it should ideally work. This leads to Section 2.3, which clearly defines the problem statement and the gaps that need to be filled. Finally, Section 2.4 introduces our proposed solution, and Section 2.5 explains the specific software model we chose for this project.

2.1 Existing System Description

A thorough examination of existing systems is a key step in software development. This analysis helps identify current features and functions in the market and reveals industry standards and best practices for design and data handling. It also uncovers limitations, such as poor emergency response integration or accessibility for elderly users. This process allows developers to learn from previous successes and failures. The evaluation focuses on elderly care and assistive technology to show the unique value of the proposed project. Benchmarking against existing products is crucial for creating a competitive system [4].

The analysis reviews ten specific applications in mobility, healthcare, and safety, focusing on both market leaders and unique assistive technologies like Practo, MyChart, Life360, and Careem. It identifies technical limitations, such as the lack of integrated AI fall detection and a unified community support network for the elderly. A comparison table summarizes the evaluated systems and highlights gaps in senior care technology.

Careem: It functions as a multipurpose mega-app that primarily facilitates lift- hailing and localized delivery services across South Asia and the Middle East. Through a centralized interface, the program provides a robust mobility structure that allows druggies to bespeak transportation and access vital community services efficiently. The system utilizes secure payment gateways and comprehensive GPS monitoring to manage high sale volumes while ensuring stoner safety. Although the platform offers some community support through its devoted care enterprise, the underpinning armature remains concentrated on marketable distribution and general- purpose transportation. Accordingly, the operation lacks the technical senior backing features and medical-specific triggers necessary to support independent elderly living. This marketable exposure creates a functional gap,

as the interface doesn't prioritize the health monitoring depth needed for senior care[5].

The main features of the Application are listed below:

- Real-time GPS vehicle tracking.
- Multi-service integration.
- Secure digital wallet payments.
- User and captain rating systems.
- Location based service discovery.

The limitations of the application are listed below:

- No integrated IoT health monitoring.
- Lacks automated accident detection.
- UI is not senior optimized.
- No non commercial volunteer network.
- Absent of medication management.

Practo: It serves as a comprehensive healthcare operation platform that connects cases with medical professionals through videotape consultations and digital appointment booking. The system acts as a digital health depository where druggies store medical records, manage conventions, and search for technical conventions throughout the region. It's specifically designed to streamline clinical executive tasks by digitizing the case- provider commerce for bettered effectiveness. still, the platform remains largely reactive, as it requires the stoner to manually initiate contact during a health occasion rather than furnishing constant background monitoring. It lacks the real- time safety shadowing and automated exigency triggers that are critical for seniors living in independent surroundings. likewise, the armature does not offer community- grounded physical backing or levy matching to support diurnal non-medical tasks[6].

The main features of the Application are listed below:

- Online doctor video consultations.
- Digital medical record storage.
- Medicine delivery and lab bookings.
- Health articles and wellness tips.
- Appointment scheduling and reminders.

The limitations of the application are listed below:

- No real time IoT sensor integration.
- No volunteer coordination mechanism.
- Lacks automated distress triggers.
- No mobility-matched navigation.

- High dependence on manual input.

Life360: It functions as a family safety and position- participating operation designed to keep family members accompanied regarding each other's precise whereabouts. Through the use of advanced geofencing technology, the system provides automated appearance cautions for specific locales and includes introductory crash discovery features for enhanced driving safety. The operation primarily targets families who wish to maintain a digital safety circle through constant real- time chart updates and accompanied communication. Despite its strengths in general position mindfulness, the system lacks technical medical integration and does not give the necessary tools for diurnal physical task backing. likewise, the absence of adaptive availability features makes it delicate for seniors with motor or visual limitations to navigate the interface singly. Accordingly, while the platform excels at tracking movement, it fails to offer the visionary health monitoring and exigency triggers needed for holistic senior care [7].

The main features of the Application are listed below:

- Real time location Circles.
- Geo fencing and place alerts.
- Driver safety and crash detection.
- Private family messaging.
- SOS Help alerts to contacts.

The limitations of the application are listed below:

- No medication or health tracking.
- Absent of AI driven fall detection.
- No community volunteer network.
- No physiotherapy or wellness tools.
- Limited senior friendly UI options.

MyChart: It functions as a case- facing gate that allows druggies to pierce their electronic health records and communicate directly with their medical providers. The operation enables druggies to view test results, manage billing, and schedule preventative care movables within a sanitarium network. It provides a largely secure terrain for sensitive health data, making it a standard in clinical settings for data operation. still, it operates primarily as an executive tool for sanitarium systems and does n't give features for diurnal life backing or social commerce. Its complexity frequently requires a high position of digital knowledge, which creates a significant hedge for senior druggies who warrant harmonious caregiver support. Accordingly, while the platform is effective for medical attestation, it fails to offer the visionary health monitoring and automated exigency triggers needed for independent living[8].

The main features of the Application are listed below:

- Direct EHR access and test results.

- Messaging with healthcare teams.
- Prescription renewal requests.
- Billing and insurance management.
- Proxy access for family caregivers.

The limitations of the application are listed below:

- No real time safety monitoring.
- Complexity of user interface.
- No community or volunteer modules.
- Lacks navigation or mobility tools.
- No proactive emergency detection.

Medisafe: It functions as a technical drug operation operation designed to help druggies cleave to complex medicine rules through customized cautions and shadowing. The operation provides vital medicine commerce warnings and allows druggies to attend their schedules with caregivers, who admit announcements if a cure is missed. It focuses heavily on the behavioral aspect of health, exercising announcement continuity to insure harmonious medical compliance. While effective for drug operation, the system lacks broader safety features. It remains a single- purpose tool that does not address the critical physical backing requirements of elderly citizens during an exigency. Accordingly, the platform fails to give the holistic safety net needed for independent living. Its narrow focus on drugstore data creates a functional gap, as it does not integrate physiological monitoring or community support modules. Eventually, the operation serves as a niche mileage rather than a comprehensive senior life- support ecosystem [9].

The main features of the Application are listed below:

- Personalized medication reminders.
- caregiver notification.
- Drug to drug interaction warnings.
- Refill reminders and stock tracking.
- Progress reports for doctors.

The limitations of the application are listed below:

- No location or GPS features.
- No emergency or SOS triggers.
- Lacks community support integration.
- No physical therapy or wellness AI.
- No real time vital sign tracking.

Be My Eyes: It is a community-driven application that connects blind and low-vision individuals with sighted volunteers through live video calls. The application serves

as a tool for visual assistance, where volunteers help users identify objects, read labels, or navigate environments. It relies entirely on a global network of altruistic volunteers and has established a successful model for peer-to-peer assistance. However, the system is strictly limited to visual tasks and does not include any automated safety features or medical tracking. While it addresses the social support gap between the users, it is not designed to handle physical emergencies like medical collapses or falls.Ultimately, it lacks the proactive health triggers and integrated hardware telemetry necessary for independent senior living [10].

The main features of the Application are listed below:

- Live video connection to volunteers.
- Global community-based assistance.
- Specialized help from company agents.
- Multi language support.
- Simplified one button interface.

The limitations of the application are listed below:

- No automated AI safety sensors.
- Lacks health or medication tools.
- No physical task assistance.
- No location or safety geofencing.
- Dependent on volunteer availability.

Snug Safety: It provides a daily check-in service for seniors who live alone, functioning as a proactive "well-being" monitor. The application requires users to tap a button at a scheduled time each day to confirm they are safe; if they fail to do so, the app alerts their emergency contacts. It is designed to be extremely simple, focusing on preventing silent accidents where someone might be incapacitated for days. However, the system is highly manual, user friendly and cannot detect emergencies in real-time between check-in windows. It lacks sophisticated sensors, AI fall detection, and medical management features, making it a very basic safety net.while the platform offers peace of mind through scheduled verification, it fails to address the dynamic and unpredictable nature of medical emergencies [11].

The main features of the Application are listed below:

- Daily automated check-in timer.
- Emergency contact notification system.
- Wellness quotes and daily motivation.
- Simple I am OK button interface.
- Basic health stats tracking.

The limitations of the application are listed below:

- No real time accident detection.
- Lacks IoT or medical integration.
- No community or volunteer modules.
- Entirely dependent on manual taps.
- No navigation or mobility support.

AARP Now: It is a lifestyle and advocacy application designed for the over 50 demographic, providing news, local event information, and membership benefits. The system offers resources on healthy living, financial planning, and social opportunities tailored to older adults. It serves as a digital hub for community engagement and benefits management. While it excels at information delivery and social networking, it is not a technical assistive tool. It lacks the safety triggers, emergency response features, and real time health monitoring necessary for high-risk elderly individuals to use the system.while the platform offers peace of mind through scheduled verification, it fails to address the dynamic and unpredictable nature of medical emergencies which create more risk for the users specially elders [12].

The main features of the Application are listed below:

- Local event and workshop discovery.
- Health and wellness articles.
- Membership discount management.
- News and advocacy updates.
- Social forum access.

The limitations of the application are listed below:

- No emergency SOS or fall detection.
- No real time medical data tracking.
- Lacks task-based volunteer matching.
- No navigation or travel escorting.
- Only Information tool.

Seeing AI: It is an AI-powered camera application designed to narrate the world for the visually impaired. It uses computer vision to read text, recognize people, and describe objects or scenes in real time. The application is a technological leader in accessibility, using on-device machine learning to provide immediate feedback. Despite its advanced visual recognition, it does not include health monitoring or social support features. It acts as a specialized sensory aid rather than a comprehensive lifestyle assistant for seniors who may have mobility or health issues beyond vision.The limitation confirms that the system remains a niche utility that does not address the multi-dimensional safety needs of independent senior living. Ultimately, the application serves a specific sensory function but fails to provide a unified wellness ecosystem [13].

The main features of the Application are listed below:

- Real-time text to speech.
- Facial recognition and emotion detection.
- Product barcode and color identifier.
- Scene and object description.
- Currency and document reading.

The limitations of the application are listed below:

- No health or medication tracking.
- No emergency or accident alerts.
- Lacks community or volunteer support.
- No GPS or navigation guidance.
- High resource consumption.

SilverSneakers Go: It is a fitness and wellness app designed with seniors in mind, providing health tracking and exercise regimens. In order to prevent falls, the system offers guided exercises that emphasize strength, mobility, and balance. Within their insurance network, it enables users to locate local fitness classes and monitor their activity. However, there is no emergency response or real-time medical monitoring offered by the app, which is solely focused on fitness. It works as a specialized fitness and wellness application designed for seniors, providing structured workout plans and activity tracking to promote healthy aging, but it cannot identify or react to a fall if it actually happens, according to Gemini. The system offers guided workouts that concentrate on mobility, strength, and balance all of which are essential for preventing falls over the long term. Its interface allows users to track their physical activity and find nearby locations[14].

The main features of the Application are listed below:

- Personalized senior fitness plans.
- Activity and progress tracking.
- Local gym and class finder.
- Guided exercise videos.
- Health and nutrition scheduling.

The limitations of the application are listed below:

- No fall detection or safety sensors.
- No medical or prescription tools.
- Lacks emergency SOS features.
- No community task assistance.
- No real time vital sign telemetry.

Table 2.1 provides a systematic comparison of the evaluated systems against the proposed project. This comparison demonstrates that while existing market leaders excel in individual domains like mobility or location tracking, they fail to provide a unified, proactive care environment.

Table 2.1: Applications Comparison

Features	Applications										
	Careem [5]	Practo [6]	Life360 [7]	MyChart [8]	Medisafe [9]	Be My Eyes [10]	Snug Safety [11]	AARP Now [12]	Seeing AI [13]	SilverSneakers Go [14]	Proposed System
AI Fall Detect	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
IoT Health	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
Volunteer Match	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗	✓
Medicine Reminder	✗	✓	✗	✓	✓	✗	✗	✗	✗	✗	✓
Voice UI	✗	✗	✗	✗	✗	✓	✗	✗	✓	✗	✓
Emergency SOS	✓	✗	✓	✗	✗	✗	✓	✗	✗	✗	✓
Geofencing	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗	✓
Wellness AI	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	✓
Social Circle	✓	✗	✓	✗	✓	✓	✗	✓	✗	✓	✓
Object ID	✗	✗	✗	✗	✗	✓	✗	✗	✓	✗	✓

2.2 Future System Usage Analysis

The conceptualization of the proposed system centers on a proactive digital environment where the Smart Assistant for Senior Citizens transforms from a passive tool into an autonomous caregiver. An operational journey will begin when a user interacts with a senior optimized dashboard that immediately presents real time biometric telemetry alongside current emergency safety status. Unlike the reactive consultation models seen in Practo, this system will utilize a constant background monitoring architecture to trigger immediate alerts without the need for manual user initiation. The platform will accomplish these tasks by processing integrated IoT sensor data, which will deliver significant efficiency gains and time savings compared to the manual check in timers of Snug Safety. By incorporating automated AI fall detection and medical distress triggers, the Smart Assistant will eliminate the dangerous dependency on manual input that currently limits the effectiveness of family safety nets like Life360. Users will also experience new capabilities through a localized non commercial volunteer network and task request modules to improve their independence without the high costs of commercial transit services like Careem. Furthermore, the modular framework of the application will ensure long term relevance by allowing the system to easily accommodate emerging wearable technologies and evolving healthcare standards. This adaptability positions the platform as a timely and relevant solution that bridges the gaps between fragmented services while proactively anticipating the future demands of an aging global population.

2.3 Problem Statement / Limitations

The central challenge faces significant challenges due to a reliance on fragmented digital tools that do not offer comprehensive safety for independent living. Current applications like Careem or Practo are limited and do not include key features such

as automated distress detection and real-time health monitoring. Seniors struggle with these tools since they often require manual interaction or advanced digital skills, which may not be available during emergencies. This issue also impacts caregivers, who are anxious about silent emergencies. These technological gaps delay emergency responses and reduce user autonomy. The project aims to develop a smart assistant that fills all these gaps through automated IoT integration and community support, promoting secure independence for users to make things easy.

This research operates within specific technical and financial constraints that define the current boundaries and primary limitation involves the high cost of the hardware, as the implementation requires near to medical grade sensors that cost approximately thirty five to thirty seven thousand per unit for the specialized wristbands. This financial requirement impacts the scalability of the initial prototype, yet the project manages this constraint by prioritizing the reliability of high fidelity biometric data over mass market affordability at this phase. Furthermore, the physiotherapy module necessitates that users activate their camera for pose correction during exercise sessions, which introduces significant privacy concerns regarding live video data. The phase also lacks the infrastructure to establish an independent volunteer network, so the architecture focuses on a robust notification system that alerts designated caregivers while facilitating connections with established humanitarian organizations such as Edhi, Chhipa, Al Khidmat, AKDN, and the UN. These limitations remain a conscious part of the development strategy, ensuring that the final deliverable remains a technically viable and realistic solution within the established research framework.

2.4 Proposed Solution

This section presents the proposed solution designed to address the critical gaps in safety and the technical limitations identified in the previous analysis of existing systems. The Smart Assistant for Senior Citizens is a sophisticated web based platform that functions as a centralized command center for geriatric care and autonomous monitoring. Unlike the fragmented and reactive models of applications like Practo or Life360, this solution shifts the operational focus toward a proactive architecture that integrates high fidelity biometric data with community support. The core of the web application is built upon a modular framework designed to specifically overcome the manual dependency problem identified in Snug Safety. By utilizing the Biometric Telemetry Module, the system communicates directly with near medical grade sensors to track vital signs in real time, which directly addresses the hardware limitation by ensuring that the thirty five to thirty seven thousand spent on each unit results in professional grade accuracy that triggers emergency alerts without requiring user input. To resolve the privacy concerns mentioned in the previous section, the Physiotherapy Module operates behind a high security layer using strict encryption protocols so that seniors perform exercises in front of a camera for pose correction while their visual data remains protected from unauthorized parties. Furthermore, the Volunteer Module solves the infrastructure limitation by providing a direct digital bridge to established organizations such as Edhi, Chhipa, Al Khidmat, AKDN, and the UN, effectively replacing the high costs of commercial services like Careem with a reliable non commercial assistance network. The value proposition of this web based system lies in its ability to centralize fragmented services into a single secure dashboard that reduces the cognitive load on elderly users and saves life saving

minutes during a crisis through the AI Fall Detection and Emergency Safety Modules. By automating these processes, the platform provides caregivers with a state of constant situational awareness and fulfills the project objectives of Section 1.3 by enhancing geriatric quality of life through technological integration and community based humanitarian support. This proposed solution establishes a scalable framework for independent living that subsequent chapters elaborate on through detailed requirements, architectural design, and technical implementation strategies.

2.5 Software Process Model

A software process model is a structured framework guiding software project development from start to finish. The V-Model outlines specific phases and workflows by linking development activities to testing activities. It helps bring order by requiring verification in the design phase and validation in the execution phase. The use of this model offers benefits like improved project planning, better resource management, and enhanced team communication. The choice of this model is based on project factors such as size, complexity, and requirements stability. The V-Model ensures early error detection following engineering standards and is selected for the Smart Assistant for Senior Citizens [16]. This choice aligns with established IEEE frameworks for software verification and validation processes to ensure a high integrity deliverable [17].

The V-Model is chosen as the main framework due to strong verification processes that meet the safety needs of the modules. It allows for integrating costly biometric hardware and ensures medical privacy. Implementation starts requirements analysis which followed by development activities linked to testing stages for project success.

2.5.1 Introduction

The V-Model is the main development framework for the Smart Assistant for Senior Citizens, focusing on the relationship between system design and functional testing. It includes a parallel testing phase for each of the twelve core modules, ensuring they meet technical requirements before moving forward. This method is crucial due to the use of expensive sensors and hardware, which require thorough verification to avoid costly failures. Acceptance criteria are set early to ensure the AI Fall Detection and Physiotherapy protocols are sound. The process involves requirement analysis, system design, and a testing phase, ensuring user safety in geriatric care. According to established engineering literature, the V-Model provides the necessary traceability to ensure that complex software-hardware interfaces remain consistent with user needs throughout the lifecycle . Table 2.1 illustrates the workflow, and the figure is placed immediately after with proper caption and source citation [18].

2.5.2 Justification

The V-Model is the best framework due to its focus on early verification, which is crucial for the twelve core modules. This is necessary for precise integration of costly biometric sensors for monitoring senior health. The structured approach pairs development with validation, reducing risks of inconsistencies. It effectively manages hardware and software interactions within a set academic timeline, meeting safety and reliability needs.

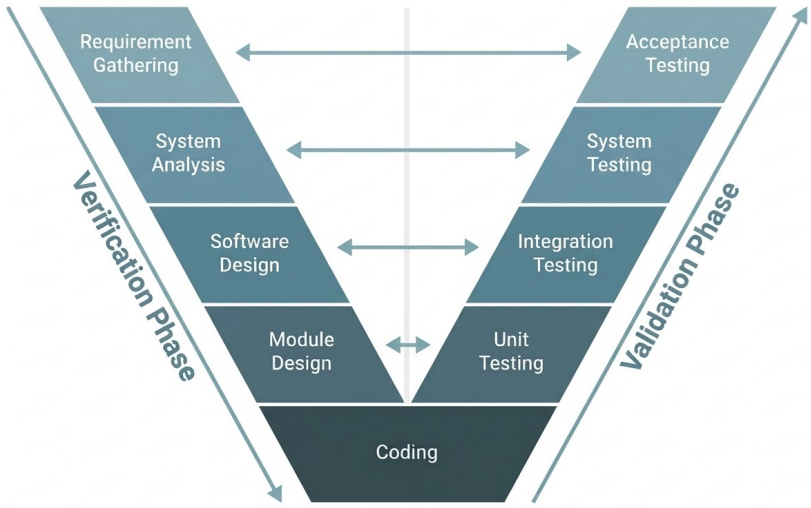

Figure 2.1: V Process Model [18]

2.5.3 Steps

This section maps the complete project development journey for the Smart Assistant for Senior Citizens using the structured V-Model framework. It provides a future-oriented roadmap that details how each architectural phase will evolve from initial conceptualization to the final validated system. By following this disciplined sequence, the development team will ensure that every one of the twelve core modules meets the rigorous safety and reliability standards required for geriatric care.

- **Requirement Analysis Phase**

The development team will start the project with a thorough requirement analysis to define the needs of senior users. They will collect data on emergency safety protocols and task request modules, ensuring all stakeholder expectations are documented. Key activities include creating a Software Requirement Specification document and acceptance criteria for the final validation stage. This early boundary setting aims to prevent scope creep and misalignment, providing a verified blueprint for design and testing.

- **System Design and Architecture Phase**

Following the requirements phase, engineers will move into system design and architecture to create a technical framework. They will design the high-level system architecture, focusing on how biometric sensors connect to the assistant's central processing unit. This System Design Documents detailing data flow, database schema, and communication protocols for real-time monitoring of users. This phase is crucial for outlining the structure of the twelve-module and will guide the coding and integration phases.

- **Module and Unit Design Phase**

Once the high-level architecture is finalized, the project will enter the detailed module and unit design phase. Developers will create low-level design specifications for individual units, like the biometric telemetry processor and the automated notification system. This includes detailed algorithms and flowcharts. The team will produce a Detailed Design Document as a technical manual. This step ensures performance and reliability for the twelve modules before unit testing begins.

- **Implementation and Coding Phase**

During the implementation and coding phase, the development team will convert design specifications into functional software, writing the source code while ensuring biometrics calibration to the software. This aims to produce the executable version of project, following coding standards from the design. The main deliverable comprises completed software units and an integrated code base for verification in the V-Model, a transition from design to functional reality and paving the way for testing and validation cycles.

- **Verification and Validation Phase**

The final stages of the project will include a verification and validation phase, executing unit, integration, and systems testing. The team will assess the performance against initial requirements. All twelve modules will undergo stress testing to verify the function effectively. Deliverables will comprise Test Summary Reports and a validated product for deployment to users. This is crucial for identifying technical discrepancies before implementation, ensuring a successful and high-integrity project completion for users.

Chapter 3

Software Requirements Specification

A Software Requirements Specification (SRS) is a formal document that defines the roadmap for a software development project, detailing system requirements and performance expectations. It captures a structured description of the system's behavior, including functional requirements, operational constraints, and quality attributes. The SRS is the foundational blueprint for design, development, and testing, fostering a shared understanding among stakeholders and providing a contractual basis for project scope and delivery benchmarks. Documentation standards are guided by IEEE 830-1998[19].

This chapter outlines the operational and technical framework of the system, starting with the project scope and the target audience. It details the functional and non-functional requirements, categorizing the latter into critical software quality attributes, performance benchmarks, and safety protocols. The chapter also describes the elicitation techniques for gathering requirements and summarizes the project development lifecycle, providing essential insights for the subsequent architectural and design specifications.

3.1 Introduction

This section outlines the Software Requirements Specification (SRS) as a critical guide for the software development lifecycle, transitioning from general concepts to detailed requirements. It focuses on delivering safe and reliable assistance for senior citizens by ensuring all development efforts align with project objectives. The SRS will consistently trace requirements from conception to validation, minimizing technical ambiguity and the potential for costly rework, while serving as a benchmark for evaluating the final product against stakeholder expectations.

3.1.1 Document Scope

The scope of this Software Requirements Specification (SRS) document outlines the functional and non-functional requirements of the proposed software platform. It details the system's logical behavior, including AI-driven fall detection algorithms, IoT data integration protocols, and user interface requirements for seniors and volunteers. The document clarifies that it does not include hardware specifications, low-level design decisions, code implementation details, or project management elements such as budgets or development schedules, providing a focused technical foundation for stakeholder expectations.

3.1.2 Audience

This Software Requirements Specification document is designed to serve a diverse group of stakeholders, each playing a distinct role in the lifecycle of the system. The document provides a centralized technical reference that ensures all participants maintain a unified vision of goals and operational limits. The following list identifies these primary audience along with the specific strategic and technical benefits they derive from the SRS.

- **Software Developers:** These are the technical engineers responsible for the system's construction, who represent the primary technical audience. They will utilize the SRS as a precise blueprint to implement the AI-driven fall detection and IoT health monitoring modules, ensuring that the code remains strictly aligned with the specified functional requirements.
- **Project Supervisors and Evaluators:** This group consists of the academic and technical authorities who oversee the development of the "Smart Assistant for Senior Citizens." They will refer to this document to assess the project's technical feasibility and will utilize the documented specifications as a formal benchmark to validate that the final deliverable meets all established academic and professional standards.
- **Quality Assurance and Testing Team:** These are the specialists focused on system reliability, safety, and performance verification. They will derive comprehensive test cases directly from the SRS and will benefit from the document by using it to verify that every functional trigger and safety constraint operates correctly before the product is deployed to the end users.
- **Stakeholders and Caregivers:** This audience includes the family members and healthcare providers who act as the primary support network for senior citizens. They will use the SRS to gain a clear understanding of the system's capabilities, and they will benefit from the detailed non-functional requirements which guarantee that the platform maintains high levels of data privacy, safety, and reliability.

3.2 Functional Requirements

Functional Requirements (FR) define the specific actions and behaviors that a software system must perform to meet user needs and achieve its goals. They outline the inputs, processes, and outputs of the system, detailing its interactions with users, hardware, and other systems. FRs are crucial in software development as they provide a clear framework for system design, implementation, and testing, ensuring all necessary features are correctly included and validated. Well-defined FRs enhance communication among stakeholders and ensure the final system aligns with user expectations. IEEE standards recommend that these requirements be documented in a structured manner for traceability throughout the development lifecycle.[20]

Table 3.1 shows a well organized and complete list of the functional requirements that have been found for the proposed system. It gives a well organized, in depth look at what the system is expected to be able to do. For a full understanding of all the functional requirements, readers should look at this table. The table is a useful tool for looking at how the system works and what it can do in the next sections.

Table 3.1: Functional Requirements

Requirement ID	Description
FR1	User can create and manage profile.
FR2	User can sign in securely.
FR3	System can track heart rate and oxygen.
FR4	System can detect physical falls.

Continued on next page

Table 3.1 – *Continued from previous page*

Requirement ID	Description
FR5	System can send SOS alerts.
FR6	User can request daily task help.
FR7	Volunteer can accept help requests.
FR8	Caregiver can track live location.
FR9	User can receive medicine reminders.
FR10	User can use voice commands.
FR11	System can identify household objects.
FR12	Admin can manage volunteer accounts.
FR13	User can follow health exercises.

3.3 Non-Functional Requirements

It define the specific criteria used to judge the operation of a system rather than its specific behaviors. While functional requirements describe what the system does, nonfunctional requirements specify how the system performs its functions, addressing essential qualities such as usability, reliability, performance, security, and maintainability. These requirements are critical to software development because they often determine the overall success or failure of a product; even a feature-rich system may be rejected if it is slow, insecure, or difficult to use. In real-world environments, it directly impact the user experience and the long term viability of the software. Consequently, they act as the structural constraints that shape the final product's architecture and operational effectiveness [21].

This section outlines the quality and operational constraints of the project, beginning with software quality attributes like usability and reliability. It specifies performance requirements for real-time data processing and details safety requirements alongside non-functional considerations such as data privacy and device compatibility. These categories serve to ensure the project upholds high standards of reliability and efficiency, delineating the technical safeguards and performance benchmarks for system success.

3.3.1 Software Quality Attributes

Software quality attributes are non-functional characteristics that define the overall effectiveness, usability, and long-term viability of the proposed system. These attributes represent the essential quality benchmarks, such as reliability and accessibility, that determine how well the software serves its users within its intended operational environment.

- **Usability:** The user interface should be intuitive and simple to navigate, featuring large icons and high contrast text to accommodate the visual needs of senior citizens.
- **Reliability:** The system should be dependable, maintaining consistent operation of the fall detection and health monitoring modules to ensure user safety at all.
- **Availability:** The application should be accessible every second, ensuring emergency alerts and assistance requests are processed without unscheduled downtime.
- **Security:** All personal health data and location information should be encrypted, only authorized caregivers and verified volunteers can access sensitive user details.
- **Maintainability:** The software architecture should be modular and well documented, efficient updates and troubleshooting without disrupting core services.

- **Portability:** The application should be compatible with various mobile operating systems, allowing software to function effectively across different devices.

3.3.2 Performance Requirements

This section details the performance expectations for the proposed system, defining the quantitative criteria necessary for deployment acceptance. Key requirements include processing IoT heart rate and oxygen data within two seconds, an AI-driven fall detection module that triggers signals within three seconds, and a mobile application that displays the main user dashboard in under five seconds. The system must also support at least fifty concurrent volunteer connections without delays in notification delivery. These criteria are essential for meeting user expectations and maintaining effective operation in critical deployment scenarios.

3.3.3 Safety Requirements

In the proposed system, mandatory constraints and operational protocols are established to prevent user harm and maintain system integrity during critical failures. It prioritizes emergency SOS alerts to notify caregivers immediately. A fall detection module will trigger safety checks upon identifying potential falls and escalate alerts if the user remains unresponsive after thirty seconds. The application uses end-to-end encryption for GPS data to prevent unauthorized access. In case of sensor disconnection, the primary emergency contact will be notified. Meeting these safety requirements is crucial for the system's trustworthy operation.

3.3.4 Other Non-Functional Requirements

The requirements are specific to the proposed system and must be addressed to ensure a complete and robust solution for the intended user base. To maintain high standards of data integrity, the shall implement a comprehensive backup protocol that secures health records every twenty-four hours. Regarding accessibility, the system must support multiple languages, including English and French, to accommodate diverse users and their support networks. The application should be optimized for low power consumption so that the battery life of the mobile device remains sufficient for a full day of continuous monitoring. Additionally, the system shall automatically notify the user when an internet connection is lost, ensuring that the senior is aware of any monitoring gaps. Adherence to these requirements contributes to the overall quality and success of the final deliverable.

3.4 Requirements Gathering Techniques Used

Requirements gathering techniques represent the processes used to collect, document, and refine the needs of stakeholders for a system. These techniques are fundamental to the software development life cycle, as they bridge between abstract user desires and concrete technical specifications. Selecting appropriate techniques directly influences the accuracy, completeness, and overall quality of the requirements captured, ensuring that the final product aligns with its intended purpose. When these are applied correctly, they minimize the risk of developing features that do not provide value to identify critical safety constraints. Consequently, the strategic application of these techniques serves as a critical role in shaping a reliable and user centric final product. These standards for elicitation

and management are essential for professional engineering practices [22].

Several techniques were utilized in the project's requirements phase, including literature reviews, stakeholder interviews, and observational studies. This multi-faceted approach aimed to balance technical AI capabilities with the accessibility needs of the elderly, allowing the project team to capture the functional needs of seniors and the operational constraints faced by caregivers.

3.4.1 Stakeholder Interviews

This document outlines a qualitative requirements gathering technique involving structured or semi-structured interviews between development teams and key stakeholders. This method is essential in software development for deeply understanding user needs, operational challenges, and overarching goals that automated tools may miss. Conducted typically in one-on-one sessions, interviews are effective in uncovering hidden requirements and clarifying ambiguities through open-ended questions. The technique aims to ensure a shared understanding of the system's purpose and validate design assumptions with end-users, helping to create a functional, user-centric product recognized as a standard elicitation practice in professional engineering frameworks[23].

During the development of the proposed system, this technique was applied through a series of sessions involving a diverse group of participants. The primary stakeholder group consisted of five senior citizens and three professional healthcare coordinators. These interviews were conducted in both residential settings and clinical environments to capture requirements in a natural context. The development team prepared a set of targeted questions focusing on daily challenges, such as the fear of falling alone, the difficulty of managing multiple medications, and the complexities of navigating modern mobile interfaces. Discussions also explored the willingness of seniors to share real time location data with volunteers and the specific types of tasks they felt most comfortable requesting through a digital platform.

The implementation of these interviews resulted in the identification of several critical functional requirements that were not initially apparent. It was learned that seniors prioritize voice activated SOS features over manual buttons due to potential mobility restrictions during an emergency. This insight directly influenced the subsequent design phase, leading to the integration of a voice-to-text command module. A significant challenge encountered was the initial skepticism regarding data privacy, which the team addressed by refining the non-functional requirements to include more robust encryption and transparent permission settings. Ultimately, these sessions provided a balanced assessment of user needs, ensuring that the final product was grounded in the actual lived experiences of its target audience.

3.4.2 Observational Studies

It involve a requirements gathering technique where the development team monitors users in their natural environment to identify how they perform tasks and interact with existing systems. This technique is essential in software development because it reveals the "silent" requirements—actions or difficulties that users often forget to mention during interviews or surveys. By observing real-world behavior, analysts can identify environmental constraints, manual workarounds, and physical limitations that directly impact

system design. This method is highly valued for providing objective data that is not filtered through the user's memory or bias, making it a critical tool for developing accessible technology. These principles of contextual inquiry are established as a standard for understanding user environments in complex software projects [24].

This technique was applied by the development team through the observation of four senior citizens in their private residences over a period of two days. The team shadowed the participants during their morning and evening routines, focusing on how they interacted with their current mobile devices and managed their daily health checks. Materials prepared for this phase included an observation checklist and a log for recording specific pain points, such as the time taken to find an app or the physical difficulty of typing on a small keyboard. The observers explored topics like the placement of the phone throughout the day and the immediate reactions of the seniors when a technology-related frustration occurred. This process was conducted with the full consent of the participants and the assistance of their primary caregivers.

The observational studies revealed key insights for the fall detection system, indicating that seniors often leave mobile devices on tables, necessitating integrated IoT sensors instead of relying solely on phone accelerometers. Standard touch screen notifications were frequently ignored due to small font sizes, leading to the need for high contrast, full screen alerts. The Hawthorne Effect posed a challenge as participants initially appeared more tech-savvy; this was mitigated by extending the observation period for users to revert to natural behaviors. Ultimately, these adjustments helped ensure the Smart Assistant's practicality in users' daily lives.

3.4.3 User Feedback

It is a crucial requirements gathering technique in software development that involves presenting existing concepts or early features to target users to assess their suitability. This method allows for direct validation of whether proposed solutions meet end-users' needs, often executed through structured walkthroughs or usability reviews. User feedback helps identify discrepancies between developers' visions and users' realities, refining requirements iteratively. This evaluative process is essential for ensuring a final product that enhances user satisfaction and adoption, establishing feedback loops as a standard for high-quality requirement validation in engineering projects[25].

During the development of the Smart Assistant for Senior Citizens, this technique was applied through a series of review sessions involving five senior citizens and two primary caregivers. These sessions were conducted in a quiet, controlled environment where the development team presented the proposed navigation flow and alert mechanisms using static design boards and interactive walk through.The participants were asked to provide their honest reactions to the layout, the clarity of the voice prompts, and the ease of requesting assistance from the volunteer network. Questions posed by the team explored topics such as the perceived helpfulness of the medication reminders and the comfort level regarding the automated fall detection alerts. The team utilized the past tense to record the specific verbal and non verbal reactions of the participants during these interactions.

The implementation of these feedback sessions resulted in several vital adjustments to the system's functional requirements. It was learned that the original terminology used for IoT Syncing was confusing for the seniors, which led to the requirement that

all technical status updates be phrased in simple, everyday language like Health Check Active. This insight directly influenced the subsequent writing of the system's notification strings. A challenge encountered during the process was the tendency for participants to be overly polite, which the team addressed by encouraging them to identify at least one thing they found difficult. Ultimately, this technique was effective in ensuring that the Smart Assistant's features were tailored to the cognitive preferences and daily routines of its target audience, contributing to a more intuitive final product.

3.5 Time Frame

A software development time frame is more than just a list of dates; it is the heartbeat of a project that keeps every phase moving in harmony. This structured schedule acts as a clear road map, guiding the team from the first spark of an idea to the final product launch. Staying on track is vital because it prevents the project from growing out of control and ensures that time and energy are spent where they matter most. By setting firm deadlines, the team maintains a steady momentum and can spot potential hurdles before they become major roadblocks. Ultimately, a well planned timeline builds trust with everyone involved, ensuring the system is ready exactly when it is needed. These core principles of organized scheduling are widely recognized as the foundation of successful engineering management [26].

Table 3.2 outlines the planned schedule for the proposed system's development. This representation provides a clear breakdown of the specific project iterations and their corresponding durations

Table 3.2: Time Frame

Phase	**Duration**
User Profile and Accessibility Management Module	30 days
Medication Scheduling and Healthcare Reminder Module	30 days
Volunteer Network and Community Support Module	30 days
Caregiver Monitoring and Location Safety Module	30 days
Voice Command and Emergency SOS Response Module	30 days
IoT Hardware Integration and Sensor Setup	30 days
AI-Powered Fall Detection and Final System Integration	30 days

References

[1] Pressman, R. S., and Maxim, B. R. (2020). Software Engineering: A Practitioner's Approach. McGraw-Hill Education.

[2] Sommerville, Software Engineering, 10th ed. Pearson, 2015.

[3] J. W. Creswell and J. D. Creswell, Research Design: Qualitative, Quantitative, and Mixed Methods Approaches, 5th ed. Thousand Oaks, CA: SAGE Publications, 2018.

[4] I. Sommerville, Software Engineering, 10th ed. Pearson Education, 2015.

[5] Careem, "Careem Care App: Community and Mobility Services," [Online]. Available: `https://help.careem.com/` [Accessed: 2-Feb-2026].

[6] Practo Technologies Pvt. Ltd., "Practo: Video Consultations and Health Records," [Online]. Available: `https://www.practo.com/` [Accessed: 2-Feb-2026].

[7] Life360, "Life360: Location Sharing and Family Safety," [Online]. Available: `https://www.life360.com/` [Accessed: 2-Feb-2026].

[8] Epic Systems Corp., "MyChart: Your Secure Health Connection," [Online]. Available: `https://www.mychart.com/` [Accessed: 30-Mar-2026].

[9] Medisafe, "Medisafe Pill Minder and Tracker," [Online]. Available: `https://www.medisafeapp.com/` [Accessed: 30-Mar-2026].

[10] Be My Eyes, "Be My Eyes - See the World Together," [Online]. Available: `https://www.bemyeyes.com/` [Accessed: 30-Mar-2026].

[11] Snug Safety, "Snug Safety: The Daily Check-in Service," [Online]. Available: `https://www.snugsafe.com/` [Accessed: 30-Mar-2026].

[12] AARP, "AARP Now App," [Online]. Available: `https://www.aarp.org/about-aarp/info-2016/aarp-now-app.html` [Accessed: 30-Mar-2026].

[13] Microsoft, "Seeing AI: Talking Camera for the Blind," [Online]. Available: `https://www.microsoft.com/en-us/ai/seeing-ai` [Accessed: 30-Mar-2026].

[14] SilverSneakers, "SilverSneakers GO Fitness App," [Online]. Available: `https://www.silversneakers.com/learn/go-app/` [Accessed: 30-Mar-2026].

[15] I. Sommerville, Software Engineering, 9th ed., Boston, MA, USA: Addison-Wesley, 2010

[16] IEEE Computer Society, "IEEE Standard for Software Verification and Validation," IEEE Std 1012-2012, vol. 1, pp. 1-223, 2012.

[17] IEEE Computer Society, "IEEE Standard for Developing Software Life Cycle Processes," IEEE Std 1074-1997, 1997.

[18] IEEE Computer Society, "IEEE Standard for Software Verification and Validation," IEEE Std 1012-2012, vol. 1, pp. 1-223, 2012.

[19] IEEE Computer Society, "IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications," IEEE, vol. 1, no. 1, 1998.

[20] IEEE "Systems and software engineering – Life cycle processes – Requirements engineering," *ISO/IEC/IEEE 29148:2018(E)*, pp. 1-104, 2018.

[21] IEEE Computer Society, "ISO/IEC/IEEE 24765:2017 Systems and software engineering – Vocabulary," *IEEE Standard*, pp. 1-541, 2017.

[22] IEEE Computer Society, "IEEE Guide for Information Technology - System Definition - Concept of Operations (ConOps)," *IEEE Std 1362-1998*, 1998.

[23] A. M. Hickey and A. M. Davis, "Elicitation technique selection: how do experts do it?," *Proceedings of the 11th IEEE International Requirements Engineering Conference*, Monterey, CA, USA, 2003, pp. 169-178.

[24] B. Nuseibeh and S. Easterbrook, "Requirements engineering: a roadmap," *Proceedings of the Conference on the Future of Software Engineering*, Limerick, Ireland, 2000, pp. 35-46.

[25] J. Nielsen, "Iterative User Interface Design," *Computer*, vol. 26, no. 11, pp. 37-41, Nov. 1993.

[26] Project Management Institute, "A Guide to the Project Management Body of Knowledge (PMBOK Guide)," 7th ed., *Project Management Institute*, 2021.

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